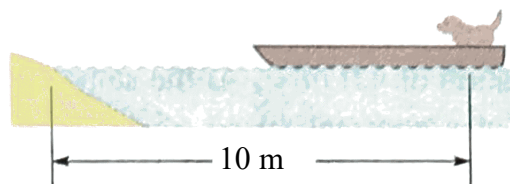


Tutorial 4: Impulse, Linear Momentum, Centre of Mass

Centre of Mass (discrete masses)

Q1.

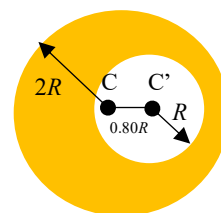
A dog, weighing 5.0 kg, is standing on a flatboat so that he is 10 m from the shore. He walks 4.0 m on the boat towards the shore and then halts. The boat weighs 20.0 kg and one can assume that there is no water resistance. How far is he from the shore at the end of this time?



[Hint: Draw out the initial and final situations and deduce the distances.]

Q2.

A uniform circular plate of radius $2R$ has a circular hole of radius R cut out of it. The centre C' of the smaller circle is a distance $0.80R$ from the centre C of the larger circle. What is the position of the centre of mass of the plate?

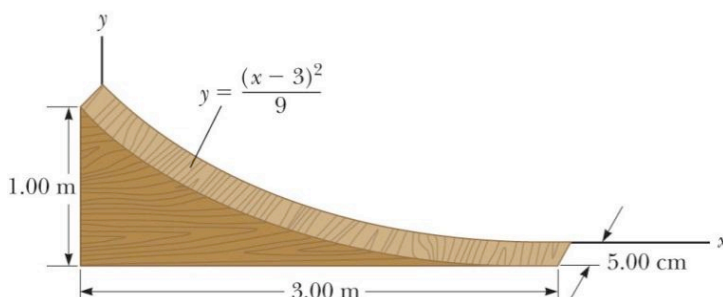


[Hint: Try subtraction.]

Centre of Mass (continuous distributions)

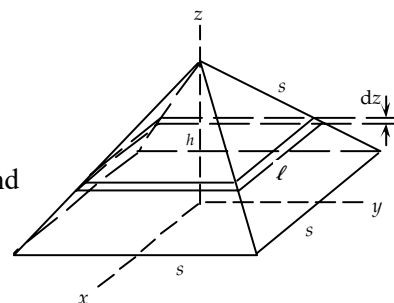
Q3.

Pat builds a track for his model car out of solid wood. The track is 5.00 cm wide, 1.00 m high, and 3.00 m long. The runway is cut so that it forms a parabola with the equation $y = (x-3)^2/9$. Locate the horizontal coordinate of the centre of mass of this track.



Q4.

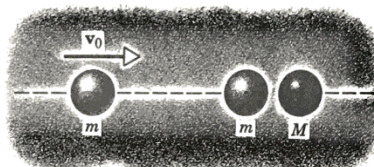
Determine the CM of a uniform pyramid that has four triangular faces and a square base with equal sides of length s .



Impulse, Conservation of Momentum

Q5.

The two masses shown on the right of the figure below are slightly separated (*just not in contact*); the left mass is incident with speed v_0 . Assuming head-on elastic collisions, (a) if $M \leq m$, show that there are two collisions and find all final velocities; (b) if $M > m$, show that there are three collisions and find all final velocities.



Q6. An alpha particle collides with an oxygen nucleus, initially at rest. The alpha particle is scattered at an angle of 64° from its initial direction of motion and the oxygen nucleus recoils at an angle of 51° on the other side of this initial direction. What is the ratio, alpha particle to nucleus, of the final speeds of these particles? The mass of the oxygen nucleus is 4 times the mass of the alpha particle.

Variable mass systems

Q7. A 70 kg astronaut floating in space in a 110 kg manned maneuvering unit (MMU) experiences an acceleration of 0.029 m s^{-2} when he fires one of the MMU's thrusters.

- (a) If the speed of the escaping N_2 gas relative to the astronaut is 490 m s^{-1} , how much gas is used by the thruster in 5.0 s?
- (b) What is the thrust produced by the thruster?

Answers

- Q1. 6.8 m
- Q2. $0.267R$ from C.
- Q3. 0.75 m
- Q4. $(x_{\text{cm}}, y_{\text{cm}}, z_{\text{cm}}) = (0, 0, s/(4\sqrt{2}))$
- Q6. 1: 3.46
- Q7. 0.053 kg, 5.2 N